

Using Python's Help System

When learning Python or any library (like Pandas, Matplotlib, NumPy), you don't need to memorize every function or method. Instead, you can use Python's built-in help tools to find exactly what you need.

1. The `help()` Function

`help()` is a built-in Python function that shows the documentation (docstring) for any Python object, function, method, or module. It provides a description, expected parameters, and sometimes examples.

`help(object)`

```
▶ ▾ help(len) # Help on built-in function len
```

```
[3] ✓ 0.0s
```

```
... Help on built-in function len in module builtins:
```

```
len(obj, /)
    Return the number of items in a container.
```

```
help(str.lower) # Help on the lower() method of strings
```

```
✓ 0.0s
```

```
Help on method_descriptor:
```

```
lower(self, /)
    Return a copy of the string converted to lowercase.
```

```
▶ ▾ help(list.append) # Help on the append() method for lists
```

```
[5] ✓ 0.0s
```

```
... Help on method_descriptor:
```

```
append(self, object, /)
    Append object to the end of the list.
```

```
import math
help(math)          # Help on the math module
```

[6] ✓ 0.0s

... Help on built-in module math:

NAME
math

DESCRIPTION
This module provides access to the mathematical functions defined by the C standard.

FUNCTIONS
acos(x, /)
Return the arc cosine (measured in radians) of x.

The result is between 0 and pi.

acosh(x, /)
Return the inverse hyperbolic cosine of x.

asin(x, /)
Return the arc sine (measured in radians) of x.

The result is between -pi/2 and pi/2.

```
import pandas as pd
help(pd.DataFrame)  # Help on the DataFrame class in pandas
```

[8] ✓ 0.2s

... Help on class DataFrame in module pandas.core.frame:

```
class DataFrame(pandas.core.generic.NDFrame, pandas.core.arraylike.OpsMixin)
| DataFrame(data=None, index: 'Axes | None' = None, columns: 'Axes | None' = None, dtype: None, copy: bool = True)
| Two-dimensional, size-mutable, potentially heterogeneous tabular data.
|
| Data structure also contains labeled axes (rows and columns).
| Arithmetic operations align on both row and column labels. Can be
| thought of as a dict-like container for Series objects. The primary
| pandas data structure.
|
| Parameters
```

2. Using Docstrings and Inline Help in Interactive Environments

If you use Jupyter Notebooks, IPython, or other interactive environments, there are even easier ways to access documentation.

a) Using ?

Append ? after a function or object to view its docstring. e.g., len?

```
> ✓ len?
11] ✓ 0.0s

.. Signature: len(obj, /)
   Docstring: Return the number of items in a container.
   Type:      builtin_function_or_method
```

b) Using Shift + Tab

Inside Jupyter, place the cursor inside the parentheses of a function and press Shift + Tab to see the function signature and a short docstring popup.

For more information always prefer using official documentation websites such as:

- <https://docs.python.org/3/>
- <https://pandas.pydata.org/pandas-docs/stable/>
- <https://matplotlib.org/stable/contents.html>
- <https://numpy.org/doc/>